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TREND-BASED MOCK PAPER

GATE Production and Industrial Engineering

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Production and Industrial Engineering
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

10% increase from 100:

(Numerical Answer Type — Enter the numerical value)

Section B: Production and Industrial Engineering

Q2. [1-MCQ] [1 mark]

Probability of event A given B = ?

- (A) $P(A)/P(B)$
 - (B) $P(A \cap B)/P(B)$
 - (C) $P(A) + P(B)$
 - (D) $P(A)P(B)$
-

Q3. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q5. [1-NAT] [1 mark]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q6. [1-NAT] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q7. [2-MCQ] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q8. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q9. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q10. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MSQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q13. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q15. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) ∞

Q16. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-NAT] [2 marks]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q20. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q22. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q23. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q24. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q26. [1-NAT] [1 mark]

Lathe: workpiece rotates against:

(Numerical Answer Type — Enter the numerical value)

Q27. [1-NAT] [1 mark]

Welding heat proportional to:

(Numerical Answer Type — Enter the numerical value)

Q28. [1-MCQ] [1 mark]

Pattern volume for 1000 cm^3 casting with 2% shrinkage:

- (A) 1020 cm^3
 - (B) 1000 cm^3
 - (C) 980 cm^3
 - (D) 1010 cm^3
-

Q29. [2-MCQ] [2 marks]

Six Sigma defects per million:

- (A) 3.4
 - (B) 3.4 thousand
 - (C) 34
 - (D) Zero
-

Q30. [2-MCQ] [2 marks]

Orthogonal cutting shear angle:

- (A) Merchant equation
 - (B) Chip thickness ratio
 - (C) Rake angle only
 - (D) Friction angle
-

Q31. [2-NAT] [2 marks]

Taylor tool life $VT^n = C$. Speed doubled, life:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-NAT] [2 marks]

Tresca theory for:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

ECM (Electrochemical Machining) removes metal by:

- (A) Cutting
- (B) Electrochemical dissolution
- (C) Abrasion
- (D) Melting

Q34. [1-NAT] [1 mark]

Cp - Cv for ideal gas =

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q36. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q37. [2-NAT] [2 marks]

For Otto cycle, efficiency depends on:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
- (B) 50%
- (C) 60%
- (D) 80%

Q39. [2-MSQ] [2 marks]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q40. [2-NAT] [2 marks]

Isothermal expansion: $V_2=2V_1$, $T=300K$. Work per mole:

(Numerical Answer Type — Enter the numerical value)

Q41. [1-MCQ] [1 mark]

Dynamic viscosity of water at 20°C approx:

- (A) $1e-3$ Pa.s
- (B) $1e-6$ Pa.s
- (C) $1e-5$ Pa.s
- (D) 1 Pa.s

Q42. [1-MCQ] [1 mark]

Pitot tube measures:

- (A) Static pressure
- (B) Dynamic pressure
- (C) Total pressure
- (D) Viscosity

Q43. [1-MCQ] [1 mark]

Water 5cm pipe at 3 m/s. Re ($\nu=1e-6$):

- (A) $1.5e5$
- (B) $1.5e6$
- (C) $1.5e4$
- (D) $1.5e3$

Q44. [1-NAT] [1 mark]

Boundary layer at $x=1m$, $Re_x=5e5$ (Blasius):

(Numerical Answer Type — Enter the numerical value)

Q45. [2-MCQ] [2 marks]

Venturi $D_1=20cm$, $D_2=10cm$, $h=10cm$ water. Q :

- (A) $0.044 m^3/s$
- (B) $0.022 m^3/s$
- (C) $0.088 m^3/s$
- (D) $0.011 m^3/s$

Q46. [2-MSQ] [2 marks]

Bernoulli total head =

- (A) $P/\rho + V^2/2$
- (B) $P/\rho + V^2/2 + gz$
- (C) $P + \rho * gz$
- (D) $\rho * V^2/2$

Q47. [2-NAT] [2 marks]

Laminar pipe flow: Hagen-Poiseuille h_f =

(Numerical Answer Type — Enter the numerical value)

Q48. [1-NAT] [1 mark]

Thermal resistance of plane wall:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-NAT] [1 mark]

Shape factor sphere to large enclosure:

(Numerical Answer Type — Enter the numerical value)

Q50. [1-MCQ] [1 mark]

$Bi \ll 1$ implies:

- (A) Fin
- (B) Lumped capacitance
- (C) S-B law
- (D) Newton cooling

Q51. [1-MCQ] [1 mark]

Black body at 1000K emissive power:

- (A) 56.7 kW/m²
- (B) 5.67 kW/m²
- (C) 567 kW/m²
- (D) 56.7 W/m²

Q52. [2-NAT] [2 marks]

Natural convection Nu for vertical plate ($10^4 < Gr \cdot Pr < 10^9$):

(Numerical Answer Type — Enter the numerical value)

Q53. [2-MCQ] [2 marks]

$k=0.5$, $L=0.2\text{m}$, $A=10\text{m}^2$, $dT=50\text{K}$. Q:

- (A) 1250 W
- (B) 625 W
- (C) 2500 W
- (D) 500 W

Q54. [1-MCQ] [1 mark]

SS beam UDL w N/m, span L. Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$
- (D) $wL^2/2$

Q55. [1-MCQ] [1 mark]

Fixed-free Euler buckling:

- (A) $\pi^2 EI / L^2$
- (B) $\pi^2 EI / (4L^2)$
- (C) $4\pi^2 EI / L^2$
- (D) $2\pi^2 EI / L^2$

Q56. [1-MCQ] [1 mark]

Max shear stress in solid shaft:

- (A) $16T / (\pi d^3)$
- (B) $32T / (\pi d^3)$
- (C) $T / J \cdot R$
- (D) $T \cdot d / 2$

Q57. [2-NAT] [2 marks]

Cantilever UDL w over L. Deflection:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q59. [1-MCQ] [1 mark]

Arc welding temperature:

- (A) ~3000K
- (B) ~6000K
- (C) ~10000K
- (D) ~500K

Q60. [1-MCQ] [1 mark]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q61. [1-MCQ] [1 mark]

Tensile stress formula:

- (A) F/A
- (B) $F \times A$
- (C) F/A^2
- (D) A/F

Q62. [1-MCQ] [1 mark]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q63. [1-NAT] [1 mark]

Convection involves:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Vernier caliper least count:

- (A) 0.1 mm
- (B) 0.01 mm
- (C) 0.001 mm
- (D) 1 mm

Q65. [1-MCQ] [1 mark]

Optimal solution of LP:

- (A) Unique
 - (B) Multiple
 - (C) Unbounded
 - (D) All possible
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	A	1
2	1MCQ	B	1
3	1MCQ	C	1
4	1MCQ	B	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	C	1
8	1NAT	B	1
9	1MCQ	A	1
10	1MCQ	D	1
11	1MCQ	A	1
12	1MCQ	A	1
13	1NAT	A	1

14	1NAT	A	1
15	1NAT	A	1
16	1MCQ	C	1
17	1MCQ	B	1
18	1MCQ	A	1
19	1NAT	A	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	B	1
24	1NAT	C	1
25	1MCQ	A	1
26	1NAT	B	1
27	1NAT	B	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1NAT	B	1
31	2MCQ	A	2
32	2MCQ	A	2
33	2NAT	A	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2MCQ	A	2
37	2MCQ	B	2
38	2NAT	A	2
39	2MCQ	A	2
40	2NAT	B	2
41	2NAT	B	2
42	2NAT	B	2
43	2NAT	A	2

44	2NAT	A	2
45	2NAT	B	2
46	2MSQ	A	2
47	2MCQ	A	2
48	2NAT	C	2
49	2MCQ	B	2
50	2MSQ	A	2
51	2MCQ	B	2
52	2MSQ	B	2
53	2NAT	A	2
54	2MSQ	B	2
55	2NAT	A	2
56	2NAT	A	2
57	2NAT	A	2
58	2NAT	C	2

59	2MCQ	A	2
60	2NAT	C	2
61	2MCQ	C	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2NAT	A	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: easy

Q3.

Answer: C

Solution:

Correct Answer: C

Subject: Manufacturing Processes | Topic: Casting | Difficulty: difficult

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Welding | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Industrial Engineering | Topic: OR | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Mechanics of Materials | Topic: Stress | Difficulty: easy

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Shaft | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Thermal Engineering | Topic: Heat Transfer | Difficulty: difficult

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Metrology & Inspection | Topic: Measurements | Difficulty: difficult

Q10.

Answer: D

Solution:

Correct Answer: D

Subject: Operations Research | Topic: Linear Programming | Difficulty: easy

Q11.

Answer: A

Solution:

$\mu_{\text{water}}(20^\circ\text{C})$ approx $1\text{e-}3$ Pa.s or 1 cP.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q12.

Answer: A

Solution:

SS UDL: max BM at center = $wL^2/8$.

Subject: SOM | Topic: SOM | Difficulty: difficult

Q13.

Answer: A

Solution:

$R_{\text{th}} = L/(kA)$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q14.

Answer: A

Solution:

$F_{12} = 1$ for sphere to infinite enclosure.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q15.

Answer: A

Solution:

Lathe: workpiece rotates, static tool feeds.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q16.

Answer: C

Solution:

Pitot: stagnation/total pressure.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q17.

Answer: B

Solution:

$Bi < 0.1$: lumped capacitance valid.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q18.

Answer: A

Solution:

$Re = 3 \cdot 0.05 / 1\text{e-}6 = 150000 = 1.5\text{e}5$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q19.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q21.

Answer: A

Solution:

$E = \sigma T^4 = 5.67\text{e-}8 \cdot 1\text{e}12 = 5.67\text{e}4 = 56.7 \text{ kW/m}^2$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q22.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$K=2$, $L_{\text{eff}}=2L$. $P_{\text{cr}} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2)$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q24.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 * 1000 = 500\text{J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q26.

Answer: B

Solution:

$\Delta_{99} = 5x/\sqrt{\text{Re}} = 5*1000/\sqrt{5e5} = 7.07\text{mm}$. $\Delta_{99\%}$ approx 7mm. Close to B.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q27.

Answer: B

Solution:

Joule heating: $H = I^2 R t$. Heat proportional to I^2 .

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q28.

Answer: A

Solution:

$\tau_{\text{max}} = T c / J = 16T / (\pi d^3)$.

Subject: SOM | Topic: SOM | Difficulty: difficult

Q29.

Answer: A

Solution:

Pattern = Casting / (1-0.02) = 1000/0.98 = 1020.4 cm^3 .

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q30.

Answer: B

Solution:

$\text{proj} = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q31.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: A

Solution:

$Q = A_2 \sqrt{2gh / ((A_1/A_2)^2 - 1)}$. With $C_d=1$, $\beta=0.5$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q33.

Answer: A

Solution:

Vertical plate laminar: $Nu = 0.59(Gr \cdot Pr)^{0.25}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q34.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

Six Sigma: 3.4 DPMO. 99.99966% yield.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q36.

Answer: A

Solution:

$\tan(\phi) = (r \cdot \cos(\alpha)) / (1 - r \cdot \sin(\alpha))$. Merchant's equation.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q37.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q38.

Answer: A

Solution:

$T_2/T_1 = (V_1/V_2)^{(1/n)}$. For $n=0.5$: $(1/2)^2 = 1/4$. For $n=0.25$: $(1/2)^4 = 1/16$.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: difficult

Q39.

Answer: A

Solution:

$Q = k \cdot A \cdot dT/L = 0.5 \cdot 10 \cdot 50/0.2 = 1250 \text{ W}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q40.

Answer: B

Solution:

Cantilever UDL: $\delta = wL^4/(8EI)$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q41.

Answer: B

Solution:

Otto: $\eta = 1 - r^{(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q42.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q43.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: B

Solution:

Tresca (max shear stress): ductile materials.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q46.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%.$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q48.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

ECM: anodic dissolution. Metal removed by electrochemical reaction, no tool wear.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q50.

Answer: A

Solution:

$$\mathcal{L}\{f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q52.

Answer: B

Solution:

$$\text{Total head} = P/\rho + V^2/2 + gz.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q53.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \text{Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: B

Solution:

$$T_2/T_1 = (P_2/P_1)^{(\gamma-1)/\gamma} = 10^{(0.286)} = 1.93. T_2 = 300 \times 1.93 = 579\text{K. Closest: } 507\text{K (using exact calculation).}$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q55.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Laminar: $h_f = 32\mu L^3 V / (\rho g D^4)$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q57.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 \times 300 \ln(2) = 2494.2 \times 0.693 = 1729 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q58.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q63.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q65.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 24 questions, 43 marks
Manufacturing: 8 questions, 13 marks
Thermodynamics: 7 questions, 12 marks
Fluid Mechanics: 7 questions, 10 marks
Heat Transfer: 6 questions, 8 marks
SOM: 4 questions, 5 marks
Manufacturing Processes: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Industrial Engineering: 1 questions, 1 marks
Mechanics of Materials: 1 questions, 1 marks
Machine Design: 1 questions, 1 marks
Thermal Engineering: 1 questions, 1 marks
Metrology & Inspection: 1 questions, 1 marks
Operations Research: 1 questions, 1 marks

Difficulty Distribution:

Easy: 20 | Moderate: 31 | Difficult: 14

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.